

Exp-27:

Implement a face detection algorithm using Open CV to detect and locate human faces in the images.

Code:

```
import cv2

import matplotlib.pyplot as plt

from google.colab import files

file = list(files.upload().keys())[0]

face = cv2.CascadeClassifier(
    cv2.data.harcascades +
    'haarcascade_frontalface_default.xml'
)

img = cv2.imread(file)

original = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

faces = face.detectMultiScale(gray, 1.1, 4)

for (x,y,w,h) in faces:
    cv2.rectangle(img,
        (x,y),
        (x+w,y+h),
        (0,0,0), 5)

output = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

plt.figure(figsize=(10,5))

plt.subplot(1,2,1)

plt.title("Original Image")

plt.imshow(original)

plt.axis('off')

plt.subplot(1,2,2)

plt.title("Face Detection")

plt.imshow(output)

plt.axis('off')

plt.show()
```

Output:

